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1 Introduction

1.1 Scope

This document describes how to reduce MUSE data with the `edps-gui` dashboard (Graphic User Interface), the recommended infrastructure to reduce data from the ESO telescope, using the ESO Data Processing System (EDPS). Details on the MUSE data reduction stream and how to configure the reduction to meet specific scientific needs are also given.

For a more extensive documentation on the `edps-gui` itself, consult the dedicated manual xxx

For a brief description of EDPS and its main concepts, see xxx.

For a detailed documentation on the MUSE pipeline itself, consult the pipeline manual available at [here](#).

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2 Reducing demo data

Follow this procedure to quickly reduce MUSE demo data. We assume that the EDPS Graphic User Interface and the MUSE pipeline and the demo data are installed in your system. For general instructions on how to install EDPS `edps-gui` the pipeline and all the needed components, please visit [this page](#).

2.1 Setting the workflow

a) blabla

1. After installation and while still in the installation virtual environment (if not, activate it [again] (`../quick/in` start the `edps-gui` dashboard by typing:

```
edps-gui
```

The `edps-gui` will start in a browser window.

```
:alt: dashboard
:name: fig_dashboard
```

2. Optionally, before starting EDPS one can specify new settings by pressing Help → Settings (see [here] (`../edps`

```
:alt: help_settings
:name: fig_help_settings
```

3. On the browser window with the `edps-gui` dashboard, press the button Start EDPS.

```
:alt: start_edps
:name: fig_start_edps
```

```
``{figure} figures/select_workflow.jpg
:alt: wkf
:name: fig-wkf
```

The EDPS Dashboard (Graphic User Interface) with the MUSE workflow loaded.

4. Choose the `muse.muse_wkf` pipeline workflow from the list in the ‘workflows’ field. The workflows offered in this selector depend on the installed pipelines. The graphic workflow representation will appear as in {numref} ‘fig-wkf’.

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2.2 Selecting the input data

- a) Press the button `Raw Data`.

```
:alt: raw_data
:name: fig_raw_data
```

- b) Press `Select Inputs`. A selection window will appear that allows to select data that are stored on a local disk.

```
:alt: select_input
:name: fig_select_input
```

- c) (Optional). Select the reduction target, configure the workflow parameter and specify the association preferences. These steps are optional. For more information see [\[here\]](#) (`configure_dataset_.md`).

- d) Press `Create Datasets`. A list of datasets appears, one line for each set of science data.

```
:alt: create_datasets
:name: fig_create_datasets
```

- e) Choose the datasets that should be processed

```
:alt: send
:name: fig_send
```

and send them to the data reduction queue by pressing `Submit to Reduction Queue`. Note that this action does not start the reduction automatically.

2.3 Start the reduction

To start the reduction of the datasets that are submitted to the reduction queue:

- a) Press `Reduction Queue`.

```
:alt: processing_queue
:name: fig_processing_queue
```

- b) (Optional). Configure the workflow and recipe parameters by pressing the wheel button `[]` (`figures/configure`) to open the configuration editor.

- c) Press the `Reduce` button. The selected data will now be processed with the default parameters (unless edited in the previous step).

```
:alt: reduce
:name: fig_reduce
```

Congratulations! You reduced your first data with `edps-gui`. All the reduced data are saved in the `EDPS_data` directory specified when executing `edps-gui` for the first time.

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3 Final products

By default, EDPS saves all the recipe products for all the executions in the directory specified at the first execution (default: \$HOME/EDPS_data). However, it is possible to save only the final products into a desired location. This can be achieved by exporting archived data reduction: press the ‘Export’ button in the ‘Archived Data’ tab (see [here] (the-reduction-archive-tab.md), or consult the [Frequently Asked Questions] (frequently-asked-questions.md).

To archive a data reduction, press the button ‘Archive’ in the ‘Reduction Queue’ tab.

Products are organized by DATASET (named as the first scientific exposure of the dataset), and TIMESTAMP (time of start of reduction)

The final products saved in the specified directory are:

- Databricks and reconstructed images of mosaics, when combination of multiple exposures has been performed. Their name format are DATACUBE_COMBINED_ and IMAGE_COMBINED_ followed by the target name (as obtained from the header keyword OBS TARG NAME of the first exposure)
- Databricks and reconstructed images of individual exposures. Their name format are DATACUBE_ and IMAGE_ followed by the exposure identifier (header keyword ARCFIELD). They are stored in the subdirectory ‘individual_exposures’

Additionally, if specified by the reduction target, also products of sky exposures (treated as regular science exposures) are saved. Their product names have the same convention as regular science exposures.

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4 Quality plots

Almost all processing tasks can display the input raw frames and the products in the so called "quality plots", which can be inspected from the `Reduction Queue` window. Those associated for the main product can be inspected by pressing the magnifying glass symbol at the right side of each dataset. To inspect those associated to each individual job (if created),

- Expand the desired dataset by pressing the black arrow on its left. The list of jobs will appear with the associated status (COMPLETED, RUNNING, PENDING)
- Press the magnifying glass symbol at the right side of the job you want to inspect. Only plots for completed jobs can be inspected.

```
:alt: reports
:name: show_reports
```